



STEAMSPACE

Tinkercad Character Figure

6th grade · ~3 hours · original theme-based characters · .STL export

Students design a small original character in Autodesk Tinkercad, keep the bounding-box volume at or under **20 cm³**, and export a single **.STL** file. On-site printing uses up to **six Bambu Lab P1S** printers; jobs may finish after the class block.

At a glance

Item	Spec
Grade	6th
Session time	~3 hours (print spill OK)
Software	Tinkercad (free class code / seats)
Size rule	$L \times W \times H \leq 20 \text{ cm}^3$
Deliverable	.STL file (design block)
Printers	6× Bambu Lab P1S (staff queue)
Theme	Changes each class — packet stays generic

Success today = design + .STL

The 3-hour block may end before prints finish on the P1S queue. Students succeed when they have a grouped solid that passes the size rule and exports as a clean .STL — not when plastic comes off the bed.

Contents: Teacher overview & arc · Classroom setup · Standards · Design tips & size rule · Student design checklist · Print-readiness sheet · Project landscape · Sources

1. Teacher overview

Students design a small original character in Tinkercad, check the bounding-box volume against the 20 cm³ rule, and export a single .STL for the staff-run P1S print queue. The packet is theme-agnostic: the teacher names the class theme (animals, heroes, robots, etc.) and students build original characters — **no franchise or copyrighted characters**.

3-hour session arc

Block	Min	Focus
Welcome & goals	10–15	What we'll make; size rule (≤ 20 cm ³); theme intro
Join Classrooms	10–15	Class code / link; verify everyone in gallery
Skills warm-up	20–25	Place / move / rotate / size / group (official Learn)
Sketch & plan	10	Thumbnail on paper: head/body/limbs; name + theme
Build 1	45–55	Blocky figure from primitives; group; check dimensions
Mid-check	10	Peer + coach: volume math, floating parts, base
Revise	25–35	Fix size, thicken thin bits, remove floaters, regroup
Export STL	15–20	Export → For 3D Print → STL; clear filename
Print-readiness handoff	10–15	Complete sheet; submit STL to queue
Share & clean	10	30-sec show; log out / leave seats tidy

If login burns time, cut the warm-up — not the size check or export.

The size audit and STL export are the two non-negotiables. Everything else flexes with group size and login friction.

2. Classroom setup (teacher)

Before the session, prepare a **Tinkercad Class** with a shareable **class code** or **class link**. Pick one join method per site:

- Google sign-on + class link
- **Class Seats** (no student email — good for field-style sessions)
- Email account + class code (typically age 13+)

Do not hardcode a STEAMSPACE-wide login procedure.

Exact account owner and join method is a per-site open question. Tell students: "Use your site's class code" and link Autodesk onboarding — don't invent a universal login.

Official help: **Teachers / Classrooms** — tinkercad.com/teachers · **Onboarding students** — tinkercad.com/help-center/onboarding-students
 · **Classrooms hub** — tinkercad.com/help-center/tinkercad-classrooms

3. Standards alignment

This packet targets **6th grade**. Official codes and wording are on the Moodle page **Arizona Standards Map** (course 45). Lead targets: **6.G.A.2** volume, **6.NS.B.3** decimal arithmetic, **6.RP.A.3.d** unit conversion, **6-8.4.a–d** design process, **6-8.6.b–c** original 3D artifact, **6.W.2.d** precise vocabulary, and **6.SL.4** short share. Grade-6 science content codes are not claimed. Re-check on the ADE Standards & Practices site before high-stakes adoption.

Band	How this activity hits it
Engineering Design (Sci)	Define a design problem; iterate on a model; optimize the solution to meet constraints (size rule)
Educational Technology	Use digital tools (Tinkercad) to create, test, and refine a 3D model; export for fabrication
Math — Geometry & Measurement	Compute bounding-box volume $L \times W \times H$; convert $\text{mm} \leftrightarrow \text{cm}$; compare to a numerical constraint ($\leq 20 \text{ cm}^3$)
ELA — Speaking & Listening	Present design choices (30-sec share); follow multi-step instructions

Honest selection rule

The Moodle Arizona Standards Map is the full code list. This packet only names the bands. Do not claim grade-6 matter, gravity, or energy content codes for this character project.

4. Design tips & common mistakes

Build tips (theme-agnostic)

- Start with a **body** box or cylinder, then head, then limbs.
- Prefer **thicker** limbs over spindly sticks (FDM printers need chunky parts).
- Keep the figure roughly **standing** with a stable base face for an easy first print.
- **Group** parts into one solid — Export turns grouped shapes into a single part.
- Add a **flat underside** if the figure should stand.
- **Original** characters only — no movie/game franchise mascots. Follow today's class theme.
- Scale down **early** if the volume creeps over 20 cm³.

Size rule — $L \times W \times H \leq 20 \text{ cm}^3$

Measure the bounding box in **mm**, convert to **cm** ($\div 10$), multiply.

Worked example

$25 \times 20 \times 40 \text{ mm} \rightarrow 2.5 \times 2.0 \times 4.0 \text{ cm} \rightarrow 20 \text{ cm}^3$ (at the limit). A cube $\sim 2.7 \text{ cm}$ on each side is about 20 cm^3 . $3 \times 3 \times 3 \text{ cm} = 27 \text{ cm}^3 \rightarrow \text{FAIL}$ — shrink or redesign.

This is a **3D volume budget** (not footprint-only) so tall figures stay small enough for a classroom "mini character."

How to check in Tinkercad

- Select the whole grouped figure.
- Read length / width / height from the inspector (confirm units are mm or cm).
- Convert all three edges to cm, multiply, compare to 20.
- Record numbers on the design checklist.

5. Worksheet — Design checklist

Name: _____ Partner: _____ Site/class: _____

Today's theme: _____

Before you build

- I joined the Tinkercad class (class code / link)
- My character is original (not a movie/game franchise mascot)
- I know the size rule: $L \times W \times H \leq 20 \text{ cm}^3$

Build checks

- I used primitives (boxes, cylinders, spheres, etc.)
- Parts that should be one object are Grouped
- No floating pieces left in the air
- Limbs / details are thick enough to see clearly (not hair-thin)
- I looked at the model from top / side / bottom

Size math (cm)

Convert if Tinkercad shows mm ($\div 10$).

Edge	Value (cm)
L (length)	
W (width)	
H (height)	
$L \times W \times H$	
$\leq 20 \text{ cm}^3$?	Yes / No

If **No**: shrink or redesign before export.

Done with design when

- Volume passes
- Peer checked my list
- Ready for print-readiness sheet + STL export

Peer checker initials: _____ Coach glance: _____

One design choice I'm proud of: _____

6. Worksheet — Print readiness (.STL)

Success today = a good .STL file (print may finish later)

Student name: _____ STL filename: _____ .stl

Character name: _____ Date/time submitted: _____

Export steps (teacher may demo)

From Autodesk Help: Export → Include (all or selection) → **For 3D Print** → **STL**.

Official: tinkercad.com/help-center/exporting-3d-models-for-3d-printing

- Everything I want is selected / included
- I exported STL (not only a screenshot)
- File is saved where the teacher asked
- Filename is clear (name + character)

Design readiness (FDM / P1S queue)

Site printers: Bambu Lab P1S (adults run the slicer). Check what you can see in Tinkercad:

- One solid / grouped figure (no loose floaters)
- Volume still $\leq 20 \text{ cm}^3$ after last edits
- No paper-thin walls or wire-thin limbs
- I have an idea for which face sits on the bed (optional): _____
- Big overhangs? Yes / No — if Yes, tell coach (supports may be needed later)

Handoff

- Design checklist complete
- STL given to teacher / upload folder
- I understand my print might finish after class

Student signature: _____ Teacher received: _____

7. Project landscape — teacher extensions

Landscape scan of 12 elementary/middle Tinkercad character-figure projects. Use for teacher choice or multi-session extensions — **not** as a replacement for Autodesk Tier A Export / Classrooms docs.

Best fits for this ~3 hr arc

Project	Why it fits	Watch-outs
Create Your Own Avatar (Tinkercad official)	No-experience entry; sketch → shapes; print-minded	~5 hr official — compress with this arc
Tinkercad Figures Template (Instructables)	Fast customize-from-template path	Community UI; keep Tier A for Export
Build a Superhero (Tinkercad tutorial)	Short guided customize; high engagement	Keep theme-agnostic / original heroes only
Chibi figure (Instructables)	Primitive spheres/cylinders; grouping practice	Style hook only — no anime IP

Stretch / multi-session (defer for this packet)

- Articulated creatures + connectors — joints/tolerances eat the 3 hr block
- Custom characters from reference images — strong 2D→3D but needs more sessions
- ASU Monster Manual — genetics tie-in; cross-curricular unit, not STL day
- Personal monuments — week-scale culminating exhibition

Primary classroom path stays this packet's 3-hour arc.

Lean on #1 (Avatar) pacing ideas inside the existing warm-up + Build 1 blocks. Treat articulated / multi-session projects as a **later** unit — success today is a grouped solid + STL.

8. Sources & credits

Tier A — Official / primary authority

- Tinkercad Learn hub — tinkercad.com/learn
- Tinkercad Teachers / Classrooms — tinkercad.com/teachers
- Exporting 3D models for 3D printing — tinkercad.com/help-center/exporting-3d-models-for-3d-printing
- Onboarding students — tinkercad.com/help-center/onboarding-students
- Learn the Moves (10-part lesson) — tinkercad.com/learn/overview/OPC41AJJKIKDWDV

Tier B — Classroom-quality secondary

- Wikipedia: Tinkercad, STL (file format), Fused filament fabrication
- Bambu Lab P1 product page — bambulab.com/en/p1
- Tinkercad webinars / educator videos — tinkercad.com/webinars

Tier C — Inspiration only

- Community STL export walkthrough (YouTube) — UI may drift; verify against Help
- Gallery character models — tinkercad.com/things — no franchise characters
- 12-project landscape compile — [research.md](#) → [wiki/curated-project-landscape.md](#)

Video links

- Tinkercad Learn — Learning center (lessons hub) — tinkercad.com/learn
- Learn the Moves (10-part official) — tinkercad.com/learn/overview/OPC41AJJKIKDWDV
- Teaching with Tinkercad — YouTube playlist — youtube.com/playlist?list=PLV6cmKvnKRs4119qeD7_nuNkHRkKNTSEi

Cite, do not invent.

Export / Classrooms steps follow Tier A Autodesk Help — no invented UI screenshots. Button names can change; use the Help article, not memorized layouts. Unverified site ops (filament/color, class-code owner) stay as open questions, not invented procedures.

STEAMSPACE Resource Builder packet for multi-site use. Generated by `generate_packet_pdf.py` (Gate 4 polish pass). Full source list: [sources/curated-sources.md](#)